RECALLING Article 28(b) of the Convention on the International Maritime Organization con cerning the functions of the Committee, RECALLING ALSO resolution MSC.268(85) by which i t adopted the International Maritime Solid Bulk Cargoes Code (hereinafter referred to as "the IMSBC Code"), which has become mandatory under chapter VI of the International Conv ention for the Safety of Life at Sea, 1974, as amended ("the Convention"), RECALLING FUR THER article VIII(b) and regulation VI/1 -1.1 of the Convention concerning amendment pro cedure for amending the IMSBC Code, HAVING CONSIDERED, at its 107th session, amendments to the IMSBC Code, proposed --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeC entre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf amendment procedure for amen ding the IMSBC Code, HAVING CONSIDERED, at its 107th session, amendments to the IMSBC Co de, proposed and circulated in accordance with article VIII(b)(i) of the Convention, 1 A DOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to the IMSBC Code, the text of which is set out in the annex to the present resolution; 2 DETERMINES, in accordance with article VIII(b)(vi)(2)(bb) of the Convention, that the said amendment s shall be deemed to have been accepted on 1 July 2024, unless prior to that date, more than one third of the Contracting Governments to the Convention or C ontracting Governme nts the combined merchant fleets of which constitute not less than 50% of the gross tonn age of the world's merchant fleet, have notified their objections to the amendments; 3 I NVITES Contracting Governments to the Convention to note that, in accordance with --- UR L: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCRes olutions/MSC.539(107).pdf 3 INVITES Contracting Governments to the Convention to note th at, in accordance with article VIII(b)(vii)(2) of the Convention, the amendments shall e nter into force on 1 January 2025 upon their acceptance in accordance with paragraph 2 a bove; 4 AGREES that Contracting Governments to the Convention may apply the aforemention ed amendments in whole or in part on a voluntary basis as from 1 January 2024; 5 REQUEST S the Secretary -General, for the purpose of article VIII(b)(v) of the Convention, to tr ansmit certified copies of the present resolution and the text of the amendments contain ed in the annex to all Contracting Governments to the Convention; and 6 REQUESTS ALSO th e Secretary-General to transmit copies of this resolution and its annex to Members of th e Organization, which are not Contracting Governments to the Convention.

3 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE --- URL: https://wwwcdn.imo.org/localresources/ en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf Convention.

3 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE ANNEX AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE AMENDMENT 07-23 (Consolidated version of the IMSBC Code) The complete text of the IMSBC Code is replaced with the following: "Table of contents S ection 1 General provisions Section 2 General loading, carriage and unloading precaution s Section 3 Safety of personnel and ship Section 4 Assessment of acceptability of consig nments for safe shipment Section 5 Trimming procedures Section 6 Methods of determining the angle of repose Section 7 Cargoes which may liquefy or undergo dynamic separation Se ction 8 Test procedures for group A cargoes Section 9 Materials possessing chemical haza rds Section 10 Carriage of solid wastes in bulk Section 11 Security provisions Section 1 2 Stowage factor conversion tables --- URL: https://wwwcdn.imo.org/localresources/en/Kno wledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf Section 10 Carriage o f solid wastes in bulk Section 11 Security provisions Section 12 Stowage factor conversi on tables Section 13 References to related information and recommendations Appendix 1 In dividual schedules of solid bulk cargoes Appendix 2 Laboratory test procedures, associat ed apparatus and standards Appendix 3 Properties of solid bulk cargoes Appendix 4 Index Appendix 5 Bulk Cargo Shipping Names in three languages (English, French and Spanish) 4 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL MA RITIME SOLID BULK CARGOES (IMSBC) CODE Non-c ommercial use only Non-c ommercial use only Sec tion 1 Gen eral provisions 1.1 Int roductory note It sh ould be noted that other int ernational and national regulations exist and that those regulations may recognize all o r part of the provisions of this Code.

In addition, port authorities and other --- URL: https://wwwcdn.imo.org/localresources/e n/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf may recognize al l or part of the provisions of this Code.

In addition, port authorities and other bodies and organizations should recognize the Co de and may use it as a basis for their storage and handling bye-laws within loading and discharge areas.

1.2 Car goes listed in this Code 1.2 .1 Typ ical cargoes currently shipped in bulk, toge ther with advice on their properties and methods of handling, are given in the schedules for individual cargoes.

However, these schedules are not exhaustive and the properties attributed to the cargoes are given only for guidance.

Consequently, before loading, it is essential to obtain current valid information from t he shipper on the physical and chemical properties of the cargoes presented for shipment .

The shipper shall provide appropriate information about the cargo to be shipped (see 4.2).

1.2.2 Whe re a solid bulk cargo is specifically listed in appendix 1 to this Code (indiv idual schedules --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf 1.2.2 Whe re a solid bulk cargo is specifically listed in appendix 1 to this Code (individual schedules for solid bulk cargoes), it shall be transported in accordance with the provisions in its schedule in addition to the provisions in sections 1 to 10 and 11.1.1 of this Code.

The master shall consider whether to consult the authorities at the ports of loading and discharge, as necessary, concerning the requirements which may be in force and applicabl e for the carriage.

1.3 Car goes not listed in this Code 1.3 .1 If a so lid cargo which is not listed in app endix 1 to this Code is proposed for carriage in bulk, the shipper shall, prior to loadi ng, provide the competent authority of the port of loading with the characteristics and properties of the cargo in accordance with section 4 of this Code.

Based on the information received, the competent authority will assess the acceptability of the cargo for safe shipment.

1.3.1.1 Whe --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf the information received, the competent auth ority will assess the acceptability of the cargo for safe shipment.

1.3.1.1 Whe n it is assessed that the solid bulk cargo proposed for carriage may present hazards as those defined by group A or B of this Code as defined in 1.7 , advice is to b e sought from the competent authorities of the port of unloading and of the flag State.

The three competent authorities will set the preliminary suitable conditions for the car riage of this cargo.

1.3.1.2 Whe n it is assessed that the solid bulk cargo proposed for carriage presents no specific hazards for transportation, the carriage of this cargo shall be authorized.

The competent authorities of the port of unloading and of the flag State shall be advise d of that authorization.

1.3.2 The c ompetent authority of the port of loading shall provide to the master a cert ificate stating the characteristics of the cargo and the required conditions for carriag e and handling of this shipment.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf characteristics of the cargo and the required conditions for carriage and handling of this shipment.

The competent authority of the port of loading shall also submit an application to the O rganization, within one year from the issue of the certificate, to incorporate this soli d bulk cargo into appendix 1 of this Code.

The format of this application shall be as outlined in 1.3.3.

5 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE Non-commercial use only 1.3 .3 For mat for the properties of cargoes not listed in this Code and conditions of the carriage Ten tative bulk cargo shipping name (in capital letters) Description (describe the cargo) Character istics (fill the following table) Phys ical properties Size Angle of repose Bulk density (kg/m3) Stow age factor (m3/t) Ha zard classification Class Subsidiary hazard(s) MHB Gro up Hazard (Clarify the hazard of carriage of the cargo) (De --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf age factor (m3/t) Ha zard classification Class Subsidiary hazard(s) MHB Group Hazard (Clarify the hazard of carriage of the cargo) (De termine the following types of require ments.

If no requirement is necessary, write “No special requirements”) Stowage and segregation Hold cleanliness Weather precautions Loading Precautions Ventilation Carriage Discharge Clean-up (Spe cify the emergency procedures for the cargo, if necessary) Emergency proce dures Spe cial emergency equipment to be carried Emergency procedures Emergency action i n the event of fire Medical first aid 6 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE Non-c omme rcial use only 1.4 App lication and implementation of this Code 1.4 .1 The provisions co ntained in this Code apply to all ships to which the SOLAS Convention, as amended, appli es and that are carrying solid bulk cargoes as defined in regulation 1-1 of part A of -- - URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MS CResolutions/MSC.539(107).pdf amended, applies and that are carrying solid bulk cargoes as defined in regulation 1-1 of part A of chapter VI of the Convention.

1.4.2 Alt hough this Code is legally treated as a mandatory instrument under the SOLAS C onvention, the following provisions of this Code remain recommendatory or informative: S ec tion 11 Security provisions (except 11.1.1); Sec tion 12 Stowage factor conversion ta bles; Sec tion 13 References to related information and recommendations; App endices oth er than appendix 1 (Individual schedules of solid bulk cargoes) and appendix 5 (Bulk Car go Shipping Names in three languages (English, French and Spanish)); and The t exts in t he sections for “Description”, “Characteristics (other than Hazard classification)”, “Ha zard” and “Emergency procedures” of individual schedules of solid bulk cargoes in append ix 1.

1.4.3 In ce rtain parts of this Code, a particular action is prescribed, but the respons ibility for carrying --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/I ndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf in appendix 1.

1.4.3 In ce rtain parts of this Code, a particular action is prescribed, but the respons ibility for carrying out the action has not been specifically assigned to any particular person.

Such responsibility may vary according to the laws and customs of different countries an d the international conventions into which these countries have entered.

For the purpose of this Code, it is not necessary to make this assignment, but only to i dentify the action itself.

It remains the prerogative of each Government to assign this responsibility.

1.5 Exem ptions and equivalent measures 1.5 .1 Whe re this Code requires that a particul ar provision for the transport of solid bulk cargoes shall be complied with, a competent authority or competent authorities (port State of departure, port State of arrival or fl ag State) may authorize any other provision by exemption if satisfied that such --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolu tions/MSC.539(107).pdf State of arrival or flag State) may authorize any other provision by exemption if satisfied that such provision is at least as effective and safe as that required by this Code.

Acceptance of an exemption authorized under this section by a competent authority not pa rty to it is subject to the discretion of that competent authority.

Accordingly, prior to any shipment covered by the exemption, the recipient of the exempt ion shall notify other competent authorities concerned.

1.5.2 A com petent authority or competent authorities which have taken the initiative wi th respect to the exemption: .1 sha ll send a copy of such exemption to the Organization , which shall bring it to the attention of the Contracting Parties to SOLAS; and .2 sha ll take action to amend this Code to include the provisions covered by the exemption, as appropriate.

1.5.3 The p eriod of validity of the exemption shall be not more than five years from th e date of --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMORe solutions/MSCResolutions/MSC.539(107).pdf appropriate.

1.5.3 The p eriod of validity of the exemption shall be not more than five years from th e date of authorization.

An exemption that is not covered under 1.5.2.2 may be renewed in accordance with the pro visions of this section.

1.5.4 A cop y of the exemption or an electronic copy thereof shall be maintained on boar d each ship transporting solid bulk cargoes in accordance with the exemption, as appropr iate.

1.5.5 Con tact information for the main designated national competent authorities concer ned is given in the separate document issued by the Organization.

7 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE Non-commercial use only 1.6 Conv entions Par ts A and B of chapter VI and part A-1 of chapter VII of the SOLAS Convention, as amended, d eal with the carriage of solid bulk cargoes and the carriage of dangerous goods in solid form in bulk, --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofI MOResolutions/MSCResolutions/MSC.539(107).pdf with the carriage of solid bulk cargoes an d the carriage of dangerous goods in solid form in bulk, respectively, and only the rele vant parts are reproduced below.

Chapt er VI Car riage of cargoes and oil fuels Part A Gen eral provisions Regu lation 1 Application 1 Unl ess expressly provided otherwise, this chapter applies to the carriage of cargoes (except liquids in bulk, gases in bulk and those aspects of carriage covered by other chapters) which, owing to their particular hazards to ships or persons on board , may require special precautions in all ships to which the present regulations apply an d in cargo ships of less than 500 gross tonnage.

However, for cargo ships of less than 500 gross tonnage, the Administration, if it consi ders that the sheltered nature and conditions of voyage are such as to render the applic ation of any specific requirements of part A or B of this chapter unreasonable or unnece ssary, may take other --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/ IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf application of any specific requir ements of part A or B of this chapter unreasonable or unnecessary, may take other effect ive measures to ensure the required safety for these ships.

2 To su pplement the provisions of parts A and B of this chapter, each Contracting Gover nment shall ensure that appropriate information on cargo and its stowage and securing is provided, specifying, in particular, precautions necessary for the safe carriage of such cargoes.

Regu lation 1-1 Definitions For t he purpose of this chapter, unless expressly provided otherwise, the following definitions shall apply: 1 IMS BC Code means the International Maritime Solid Bulk Cargoes (IMSBC) Code adopted by the Maritime Safety Committee of the Organization by resolution MSC.268(85), as may be amended by the Organization, provided that such amendments are adopted, brought into force and take effect in accordance with the provisions --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Indexof IMOResolutions/MSCResolutions/MSC.539(107).pdf provided that such amendments are adopted , brought into force and take effect in accordance with the provisions of article VIII o f the present Convention concerning the amendment procedures applicable to the annex oth er than chapter I.

2 Soli d bulk cargo means any cargo, other than liquid or gas, consisting of a combinati on of particles, granules or any larger pieces of material generally uniform in composit ion, which is loaded directly into the cargo spaces of a ship without any intermediate f orm of containment.

8 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE Non-c ommercial use only Reg ulation 1-2 Requir ements for the carriage of solid bulk cargoes other than grain The carriage of solid bul k cargoes other than grain shall be in compliance with the relevant provisions of the IM SBC Code.

Regu lation 2 Cargo information 1 The s --- URL: https://wwwcdn.imo.org/localresources/e n/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf IMSBC Code.

Regu lation 2 Cargo information 1 The s hipper shall provide the master or their represe ntative with appropriate information on the cargo sufficiently in advance of loading to enable the precautions which may be necessary for proper stowage and safe carriage of th e cargo to be put into effect.

Such information shall be confirmed in writing and by appropriate shipping documents pri or to loading the cargo on the ship.

2 The c argo information shall include: .1 in th e case of general cargo, and of cargo c arried in cargo units, a general description of the cargo, the gross mass of the cargo o r of the cargo units, and any relevant special properties of the cargo.

For the purpose of this regulation the cargo information required in sub-chapter 1.9 of the Code of Safe Practice for Cargo Stowage and Securing, adopted by the Organization by resolution A.714(17), as may be amended, --- URL: https://wwwcdn.imo.org/localresources/

en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf for Cargo Stowa ge and Securing, adopted by the Organization by resolution A.714(17), as may be amended, shall be provided.

Any such amendment to sub-chapter 1.9 shall be adopted, brought into force and take effe ct in accordance with the provisions of article VIII of the present Convention concernin g the amendment procedures applicable to the annex other than chapter I; .2 in th e case of solid bulk cargo, information as required by section 4 of the IMSBC Code.

3 Pri or to loading cargo units on board ships, the shipper shall ensure that the gross mass of such units is in accordance with the gross mass declared on the shipping documen ts.

4 In th e case of cargo carried in a container, except for containers carried on a chass is or a trailer when such containers are driven on or off a ro-ro ship engaged in short international voyages as defined in regulation III/3, the gross mass according to paragr aph 2.1 of this regulation shall be verified by the shipper, either by: .1 wei --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolu tions/MSC.539(107).pdf the gross mass according to paragraph 2.1 of this regulation shal l be verified by the shipper, either by: .1 wei ghing the packed container using calibra ted and certified equipment; or .2 wei ghing all packages and cargo items, including the mass of pallets, dunnage and other securing material to be packed in the container and a dding the tare mass of the container to the sum of the single masses, using a certified method approved by the competent authority of the State in which packing of the containe r was completed.

5 The s hipper of a container shall ensure the verified gross mass is stated in the ship ping document.

The shipping document shall be: .1 sig ned by a person duly authorized by the shipper; a nd .2 sub mitted to the master or their representative and to the terminal representativ e sufficiently in advance, as required by the master or their representative, to be used in the preparation of the ship stowage plan.

9 RESOLUTION MSC.539(107) (adopted on 8 June 2023) --- URL: https://wwwcdn.imo.org/local resources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf 9 RES OLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL MARIT IME SOLID BULK CARGOES (IMSBC) CODE Non-commercial use only 6 If t he shipping document, with regard to a packed container, does not provide the verified gross mass and the mast er or their representative and the terminal representative have not obtained the verifie d gross mass of the packed container, it shall not be loaded on to the ship.

Regu lation 3 Oxygen analysis and gas detection equipment 1 Whe n transporting a solid b ulk cargo which is liable to emit a toxic or flammable gas, or cause oxygen depletion in the cargo space, an appropriate instrument for measuring the concentration of gas or oxy gen in the air shall be provided together with detailed instructions for its use.

Such an instrument shall be to the satisfaction of the Administration.

2 The A dministration shall take steps to ensure that crews of ships are trained in the use of such instruments.

Regu lation 4 --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofI MOResolutions/MSCResolutions/MSC.539(107).pdf the Administration.

2 The A dministration shall take steps to ensure that crews of ships are trained in the use of such instruments.

Regu lation 4 The use of pesticides in ships App ropriate precautions shall be taken in the use of pesticides in ships, in particular for the purposes of fumigation.

Regu lation 5 Stowage and securing 1 Car go, cargo units and cargo transport units carri ed on or under deck shall be so loaded, stowed and secured as to prevent as far as is pr acticable, throughout the voyage, damage or hazard to the ship and the persons on board, and loss of cargo overboard.

2 Car go, cargo units and cargo transport units shall be so packed and secured within th e unit as to prevent, throughout the voyage, damage or hazard to the ship and the person s on board.

3 App ropriate precautions shall be taken during loading and transport of heavy cargoes or cargoes with --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Indexo fIMOResolutions/MSCResolutions/MSC.539(107).pdf 3 App ropriate precautions shall be take n during loading and transport of heavy cargoes or cargoes with abnormal physical dimens

ions to ensure that no structural damage to the ship occurs and to maintain adequate sta bility throughout the voyage.

4 App ropriate precautions shall be taken during loading and transport of cargo units an d cargo transport units on board ro-ro ships, especially with regard to the securing arr angements on board such ships and on the cargo units and cargo transport units and with regard to the strength of the securing points and lashings.

5 Fre ight containers shall not be loaded to more than the maximum gross weight indicate d on the Safety Approval Plate under the International Convention for Safe Containers (C SC), as amended.

6 All c argoes, other than solid and liquid bulk cargoes, cargo units and cargo transpor t units, shall be loaded, stowed and secured throughout the voyage in accordance with th e Cargo Securing Manual approved by the 10 --- URL: https://wwwcdn.imo.org/localresource s/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf stowed and se cured throughout the voyage in accordance with the Cargo Securing Manual approved by the 10 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE Non-c ommercial use only Adm inistration.

In ships with ro-ro spaces, as defined in regulation II-2/3.41, all securing of such car goes, cargo units, and cargo transport units, in accordance with the Cargo Securing Manu al, shall be completed before the ship leaves the berth.

The Cargo Securing Manual shall be drawn up to a standard at least equivalent to relevan t guidelines developed by the Organization.

Part B Spec ial provisions for solid bulk cargoes Regu lation 6 Acceptability for shipme nt 1 Pri or to loading a solid bulk cargo, the master shall be in possession of comprehe nsive information on the ship’s stability and on the distribution of cargo for the stand ard loading conditions.

The method of providing such --- URL: https://wwwcdn.imo.org/localresources/en/Knowledge Centre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf ship’s stability and on the distribution of cargo for the standard loading conditions.

The method of providing such information shall be to the satisfaction of the Administrat ion.

Regu lation 7 Loading, unloading and stowage of solid bulk cargoes 1 For t he purpose of this regulation, terminal representative means a person appointed by the terminal or oth er facility, where the ship is loading or unloading, who has responsibility for operatio ns conducted by that terminal or facility with regard to the particular ship.

To enable the master to prevent excessive stresses in the ship’s structure, the ship sha ll be provided with a booklet, which shall be written in a language with which the ship’ s officers responsible for cargo operations are familiar.

If this language is not English, the ship shall be provided with a booklet written also in the English language.

The booklet shall, as a minimum, include: .1 sta bility data, as required by regulation II-1/5-1; .2 bal --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Index ofIMOResolutions/MSCResolutions/MSC.539(107).pdf booklet shall, as a minimum, include: . 1 sta bility data, as required by regulation II-1/5-1; .2 bal lasting and deballasting r ates and capacities; .3 max imum allowable load per unit surface area of the tank top pl ating; .4 max imum allowable load per hold; .5 gen eral loading and unloading instructio ns with regard to the strength of the ship’s structure including any limitations on the most adverse operating conditions during loading, unloading, ballasting opera - tions an d the voyage; .6 any s pecial restrictions such as limitations on the most adverse opera ting conditions imposed by the Administration or organization recognized by it, if appli cable; and .7 whe re strength calculations are required, maximum permissible forces and moments on the ship’s hull during loading, unloading and the voyage.

3 Bef ore a solid bulk cargo is loaded or unloaded, the master and the terminal represen tative shall agree on a --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentr e/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf 3 Bef ore a solid bulk cargo is loaded or unloaded, the master and the terminal representative shall agree on a plan whi ch shall ensure that the permissible forces and moments on the ship are not exceeded dur ing loading or unloading, and shall include the sequence, quantity and rate of loading o r unloading, taking into consideration the speed of loading or unloading, the number of pours and the deballasting or ballasting capability of the ship.

The plan and any subsequent amendments thereto shall be lodged with the appropriate auth

ority of the port State.

11 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE Non-commercial use only 4 The m aster and termi nal representative shall ensure that loading and unloading operations are conducted in a ccordance with the agreed plan.

5 If du ring loading or unloading any of the limits of the ship referred to in paragraph 2 are exceeded or are --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf accordance with the agreed plan.

5 If du ring loading or unloading any of the limits of the ship referred to in paragraph 2 are exceeded or are likely to become so if the loading or unloading continues, the mas ter has the right to suspend operation and the obligation to notify accordingly the appr opriate authority of the port State with which the plan has been lodged.

The master and the terminal representative shall ensure that corrective action is taken.

When unloading cargo, the master and terminal representative shall ensure that the unloa ding method does not damage the ship's structure.

6 The m aster shall ensure that ship's personnel continuously monitor cargo operations.

Where possible, the ship's draught shall be checked regularly during loading or unloadin g to confirm the tonnage figures supplied.

Each draught and tonnage observation shall be recorded in a cargo logbook.

If significant deviations from the --- URL: https://wwwcdn.imo.org/localresources/en/Kno wledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf Each draught and tonn age observation shall be recorded in a cargo logbook.

If significant deviations from the agreed plan are detected, cargo or ballast operations or both shall be adjusted to ensure that the deviations are corrected.

Chapt er VII Car riage of dangerous goods Part A-1 Car riage of dangerous goods in solid form in bulk Regu lation 7 Definitions Dan gerous goods in solid form in bulk means any material, other than liquid or gas, consisting of a combination of particles, granules o r any larger pieces of material, generally uniform in composition, which is covered by t he IMDG Code and is loaded directly into the cargo spaces of a ship without any intermed iate form of containment, and includes such materials loaded in a barge on a barge-carry ing ship.

Regu lation 7-1 Application 1 Unl ess expressly provided otherwise, this part applies to the carriage of dangerous goods in solid form in bulk --- URL: https://wwwcdn.imo.org/lo calresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf Re gu lation 7-1 Application 1 Unl ess expressly provided otherwise, this part applies to t he carriage of dangerous goods in solid form in bulk in all ships to which the present r egulations apply and in cargo ships of less than 500 gross tonnage.

2 The c arriage of dangerous goods in solid form in bulk is prohibited except in accorda nce with the provisions of this part.

3 To su pplement the provisions of this part, each Contracting Government shall issue, o r cause to be issued, instructions on emergency response and medical first aid relevant to incidents involving dangerous goods in solid form in bulk, taking into account the gu idelines developed by the Organization.

12 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE Non-c ommercial use only Reg ulation 7-2 Docume nts 1 In al l documents relating to the carriage of dangerous goods in solid form in bul k by sea, the bulk cargo --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCent re/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf Reg ulation 7-2 Documents 1 In al l documents relating to the carriage of dangerous goods in solid form in bulk by sea, the bulk cargo shipping name of the goods shall be used (trade names alone shall not be used).

2 Eac h ship carrying dangerous goods in solid form in bulk shall have a special list or manifest setting forth the dangerous goods on board and the location thereof.

A detailed stowage plan, which identifies by class and sets out the location of all dang erous goods on board, may be used in place of such a special list or manifest.

A copy of one of these documents shall be made available before departure to the person or organization designated by the port State authority.

Regu lation 7-3 Stowage and segregation requirements 1 Dan gerous goods in solid form in

bulk shall be loaded and stowed safely and appropriately in accordance with the nature o
f the goods.

Incompatible goods shall be segregated from one another.

2 Dan --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolu
tions/MSCResolutions/MSC.539(107).pdf the nature of the goods.

Incompatible goods shall be segregated from one another.

2 Dan gerous goods in solid form in bulk, which are liable to spontaneous heating or com
bustion, shall not be carried unless adequate precautions have been taken to minimize th
e likelihood of the outbreak of fire.

3 Dan gerous goods in solid form in bulk, which give off dangerous vapours, shall be sto
wed in a well ventilated cargo space.

Regu lation 7-4 Reporting of incidents involving dangerous goods 1 Whe n an incident tak
es place involving the loss or likely loss overboard of dangerous goods in solid form in
bulk into the sea, the master, or other person having charge of the ship, shall report t
he particulars of such an incident without delay and to the fullest extent possible to t
he nearest coastal State.

The report shall be drawn up based on general principles and guidelines developed by the
Organization.

2 In th --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOReso
lutions/MSCResolutions/MSC.539(107).pdf based on general principles and guidelines devel
oped by the Organization.

2 In th e event of the ship referred to in paragraph 1 being abandoned, or in the event
of a report from such a ship being incomplete or unobtainable, the company, as defined i
n regulation IX/1.2, shall, to the fullest extent possible, assume the obligations place
d upon the master by this regulation.

Regu lation 7-5 Requirements for the carriage of dangerous goods in solid form in bulk T
he carriage of dangerous goods in solid form in bulk shall be in compliance with the rel
evant provisions of the IMSBC Code, as defined in regulation VI/1-1.1.

1.7 Defin itions For t he purpose of this Code, unless expressly provided otherwise, the
following definitions shall apply: Angl e of repose means the maximum slope angle of non
-cohesive (i.e.

free-flowing) granular material.

It is measured as the angle between a horizontal plane and the cone slope of such materi
al.

13 --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutio
ns/MSCResolutions/MSC.539(107).pdf material.

It is measured as the angle between a horizontal plane and the cone slope of such materi
al.

13 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL
MARITIME SOLID BULK CARGOES (IMSBC) CODE Non-commercial use only Ang le of repose Bul k
Cargo Shipping Name (BCSN) identifies a bulk cargo during transport by sea.

When a cargo is listed in this Code, the Bulk Cargo Shipping Name of the cargo is identi
fied by capital letters in the individual schedules or in the index.

When a cargo is dangerous goods as defined in the IMDG Code, as defined in regulation VI
I/1.1 of the SOLAS Convention, refer to 4.1.1.

Bulk d ensity means the weight of solids, air and water per unit volume.

Bulk density is expressed in kilograms per cubic metre (kg/m3), in g eneral.

The void spaces in the cargo may be filled with air and water.

Car go space means any space in a ship designated for carriage of cargoes.

Car --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResoluti
ons/MSCResolutions/MSC.539(107).pdf and water.

Car go space means any space in a ship designated for carriage of cargoes.

Car goes which may liquefy means cargoes which contain a certain proportion of fine part
icles and a certain amount of moisture.

They may liquefy if shipped with a moisture content in excess of their transportable moi
sture limit.

Car goes which may undergo dynamic separation means cargoes which contain a certain prop ortion of fine particles and a certain amount of moisture, and may undergo dynamic separ ation if shipped at a moisture content in excess of their transportable moisture limit.

Coh esive material means materials other than non-cohesive materials.

Com petent authority means any national regulatory body or authority designated or other wise recognized as such for any purpose in connection with this Code.

The competent authority shall operate independently from the shipper.

Con centrates means materials obtained from a natural ore by a process of enrichment or beneficiation --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofI MOResolutions/MSCResolutions/MSC.539(107).pdf operate independently from the shipper.

Con centrates means materials obtained from a natural ore by a process of enrichment or beneficiation by physical or chemical separation and removal of unwanted constituents.

Con signment means a solid bulk cargo presented by a shipper for transport.

Dyna mic separation means the phenomenon of forming a liquid slurry (water and fine soli ds) above the solid material, resulting in a free surface effect which may significantly affect the ship's stability.

Flo w moisture point means the percentage moisture content (wet mass basis) at which a f low state develops under the prescribed method of test in a representative sample of the material (see paragraph 1 of appendix 2).

Flow state means a state occurring when a mass of granular material is saturated with li quid to an extent that, under the influence of prevailing external forces such as vibrat ion, impaction or ship's --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCent re/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf extent that, under the influenc e of prevailing external forces such as vibration, impaction or ship's motion, it loses its internal shear strength and behaves as a liquid.

GHS me ans the ninth revised edition of the Globally Harmonized System of Classification and Labelling of Chemicals, published by the United Nations as document ST/SG/AC.10/30/R ev.9.

Grou p A consists of cargoes which possess a hazard due to moisture that may result in l iquefaction or dynamic separation if shipped at a moisture content in excess of their tr ansportable moisture limit.

Grou p B consists of cargoes which possess a chemical hazard which could give rise to a dangerous situation on a ship.

Grou p C consists of cargoes which are classified as neither group A nor group B.

14 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE Non-c ommercial use only High --- URL: https:// wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MS C.539(107).pdf 2023 AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE Non-c ommercial use only High -density solid bulk cargo means a solid bulk cargo wi th a stowage factor of 0.56 m3/t or l ess.

IMDG Code means the International Maritime Dangerous Goods (IMDG) Code adopted by the Ma ritime Safety Committee of the Organization by resolution MSC.122(75), as may be amended by the Organization.

Inc ompatible materials means materials that may react dangerously when mixed.

They are subject to the segregation requirements of 9.3 and the schedules for individual cargoes classified in group B.

Int ernational Ship and Port Facility Security (ISPS) Code means the International Code for the Security of Ships and of Port Facilities, consisting of Part A (the provisions o f which shall be treated as mandatory) and Part B (the provisions of which shall be trea ted as recommendatory), as adopted, --- URL: https://wwwcdn.imo.org/localresources/en/Kn owledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.539(107).pdf as mandatory) and Pa rt B (the provisions of which shall be treated as recommendatory), as adopted, on 12 Dec ember 2002, by resolution 2 of the Conference of Contracting Governments to the Internat ional Convention for the Safety of Life at Sea, 1974, as may be amended by the Organizat ion.

Man ual of Tests and Criteria means the seventh revised edition of the United Nations pu blication entitled Manual of Tests and Criteria (ST/SG/AC.10/11/Rev.7 and Amendment 1).

Mat erials hazardous only in bulk (MHB) means materials which may possess chemical hazar

ds when carried in bulk other than materials classified as dangerous goods in the IMDG C
ode.

Moi sture content means that portion of a representative sample consisting of water, ice
or other liquid expressed as a percentage of the total wet mass of that sample.

Moi sture migration means the movement of moisture contained in a cargo by settling and
--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/
MSCResolutions/MSC.539(107).pdf Moi sture migration means the movement of moisture conta
ined in a cargo by settling and consolidation of the cargo due to vibration and ship's m
otion.

Water is progressively displaced, which may result in some portions or all of the cargo
developing a flow state.

Non -cohesive material means dry materials that readily shift due to sliding during tran
sport, as listed in appendix 3, paragraph 1, "Properties of dry bulk cargoes".

Pot ential sources of ignition means, but is not limited to, open fires, machinery exhau
sts, galley uptakes, electrical outlets and electrical equipment unless they are of cert
ified safe type.

Rep resentative test sample means a sample of sufficient quantity for the purpose of tes
ting the physical and chemical properties of the consignment to meet specified requireme
nts.

Ship per means any person by whom or in whose name, or on whose behalf, a contract of ca
rriage --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResol
utions/MSCResolutions/MSC.539(107).pdf Ship per means any person by whom or in whose nam
e, or on whose behalf, a contract of carriage of goods by sea has been concluded with a
carrier, or any person by whom or in whose name, or on whose behalf, the goods are actua
lly delivered to the carrier in relation to the contract of carriage by sea.

Sol id bulk cargo means any cargo, other than a liquid or a gas, consisting of a combina
tion of particles, granules or any larger pieces of material generally uniform in compos
ition which is loaded directly into the cargo spaces of a ship without any intermediate
form of containment.

Sour ces of heat means heated ship structures, where the surface temperature is liable t
o exceed 55ºC.

Examples of such heated structures are steam pipes, heating coils, top or side walls of
heated fuel and cargo tanks, and bulkheads of machinery spaces.

Stow age factor means the figure which expresses the number of cubic metres which one to
nne of cargo will occupy.

Tra --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResoluti
ons/MSCResolutions/MSC.539(107).pdf Stow age factor means the figure which expresses the
number of cubic metres which one tonne of cargo will occupy.

Tra nsportable moisture limit (TML) of a group A cargo means the maximum moisture conten
t of the cargo which is considered safe for carriage in ships not complying with the spe
cial provisions of 7 .3.2.

It is determined by the test procedures, approved by a competent authority, such as thos
e specified in paragraph 1 of appendix 2.

15 RESOLUTION MSC.539(107) (adopted on 8 June 2023) 2023 AMENDMENTS TO THE INTERNATIONAL
MARITIME SOLID BULK CARGOES (IMSBC) CODE Non-commercial use only Trim ming means any lev
elling of a cargo within a cargo space, either partial or total.

Vent ilation means exchange of air from outside to inside a cargo space.

.1 Cont inuous ventilation means ventilation that is operating at all times.

.2 Mech anical ventilation means power-generated ventilation.

.3 Nat ural ventilation means ventilation that is not power-generated.

.4 Sur --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResol
utions/MSCResolutions/MSC.539(107).pdf .2 Mech anical ventilation means power-generated
ventilation.

.3 Nat ural ventilation means ventilation that is not power-generated.

.4 Sur face ventilation means ventilation of the space above the cargo.

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MARITIME SOLID BULK CARGOES (IMSBC) CODE --- URL: https://wwwcdn.imo.org/localresources/

en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.546(107).pdf RESOLUTION MSC.
546(107) (adopted on 8 June 2023) AMENDMENTS TO THE CODE FOR THE CONSTRUCTION AND EQUIPM
ENT OF MOBILE OFFSHORE DRILLING UNITS, 1989 (1989 MODU CODE) MSC 107/20/Add.1 Annex 23,
page 1 I:\MSC\107\MSC 107/20/Add.1.docx ANNEX 23 RESOLUTION MSC.546(107) (adopted on 8 J
une 2023) AMENDMENTS TO THE CODE FOR THE CONSTRUCTION AND EQUIPMENT OF MOBILE OFFSHORE D
RILLING UNITS, 1989 (1989 MODU CODE) THE MARITIME SAFETY COMMITTEE, RECALLING Article 28
(b) of the Convention on the International Maritime Organization concerning the function
s of the Committee, NOTING that mobile offshore drilling units continue to be moved and
operated internationally, RECALLING that the Assembly, when adopting the Code for the Co
nst